



MOHAMMED JAMAI

Mechatronics Engineer — Model-Based Design, Embedded Systems & Power Electronics

Toulouse, France (open to relocation) | jamai-mohammed@hotmail.com | +33 7 58 65 52 72 | linkedin.com/in/mohammed-jamai

Mechatronics engineer with 3+ years of experience in electric powertrains, power converters and model-based validation (MiL/SiL/HiL) for major automotive OEMs. Strong background in functional safety (ISO 26262), embedded C/C++ and battery modeling.

EDUCATION

ENSEEIHT – Toulouse INP

2019 – 2022 — Engineering

Degree

Electrical Engineering & Mechatronics

Université Paul Sabatier (UT3)

2021 – 2022 — Research Master's

(M2), double degree

Energy, Materials & Sustainable Development

INSSET – Université de Picardie

2018 – 2019 — Bachelor's Degree

(L3)

Embedded Systems

École Supérieure de Technologie

2016 – 2018 — DUT (2-year

technical degree)

Electronics & Industrial Computing

TECHNICAL TOOLS

Environments:

MATLAB, Simulink, Ansys, dSPACE (SCALEXIO), INCA, Polaron, Cygwin, VS Code.

Languages & Protocols:

C/C++, Python, LabVIEW, VHDL, CAN bus, SysML.

Methods & Standards:

ISO 26262, FMEA, MiL/SiL/HiL.

SOFT SKILLS

- Analytical rigor and attention to detail
- Adaptability in agile environments
- Cross-functional teamwork
- Risk-driven project coordination

INTERESTS

Running, Reading, Technological Innovation.

CORE CAPABILITIES

- Digital Twins:** high-fidelity modeling of electrified systems and architectures in MATLAB/Simulink.
- Model-Based Design & Validation:** end-to-end verification pipelines across MiL, SiL and HiL.
- Powertrain & Machines:** electric powertrains, power electronics converters and rotating machines.
- Energy Storage:** lithium-ion, redox-flow and hydrogen systems — aging, degradation and state-of-health modeling.
- Embedded Software:** C/C++ firmware, integrated diagnostics, functional safety (ISO 26262) and FMEA.

PROFESSIONAL EXPERIENCE

VITESCO TECHNOLOGIES | Model-Based Design (MBD) Engineer

May 2023 – September 2025 | Toulouse, France — Clients: Renault, Mercedes-Benz, Ford

- Modeling & Simulation:** developed MiL/HiL physical validation models in MATLAB/Simulink for power converters (PFC, DC/DC), including EMC/EMI considerations.
- Hardware Validation:** ran real-time tests and validated FPGA configurations on dSPACE SCALEXIO test benches.
- Technical Coordination:** tracked milestones and risks, and coordinated deliveries with international integration teams.

SAFRA | Embedded Software & Functional Safety Engineer

October 2022 – April 2023 | Albi, France — Project: Zero-Emission Hydrogen Fleet

- Functional Safety:** managed software requirements traceability in Polaron (ISO 26262); ran FMEA to derive technical safety requirements (TSR) from functional safety requirements (FSR) according to ASIL targets.
- Embedded Systems:** developed C/C++ control logic for thermal, hydraulic and electrical vehicle subsystems; configured CAN networks and performed pre-validation under Cygwin.

ARTELIA GROUP | Junior Energy Storage & Battery Modeling Engineer

March 2022 – September 2022 | Lyon, France — Client: TotalEnergies

- Predictive Modeling:** built electrical and mathematical models of lithium-ion and redox-flow cell degradation, aging and state of health (SoH).
- Decision Support:** developed techno-economic tools (CAPEX/OPEX) to optimize grid-scale storage deployment.

INTERNSHIPS & TECHNICAL PROJECTS

SAFRAN & ARIANEGROUP | Advanced Engineering Project

2022 (6 months)

- Designed the architecture of a turboramjet for an electric aircraft; assessed electromagnetic coupling, mechanical stress limits and power-electronics control interfaces.

CNRS | Research Laboratory Internship

2021 (3 months)

- Designed and built a high-precision data acquisition (DAQ) PCB to measure, in real time, the thermal and mechanical strain behavior of shape-memory alloys under stress testing.